Wave Resistance of Two Systematic Yacht Hulls by a Neumann–Kelvin Panel Method

Verification report on the `wave-resistance` code.

Repository: `https://github.com/ignaziomviola/wave-resistance`
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Abstract

The `wave-resistance` code solves the Neumann–Kelvin problem for a bare sailing-yacht canoe body by a constant-source panel method, with the Delft Systematic Yacht Hull Series (DSYHS) hulls Sysser 01 and Sysser 50 as the worked cases. This report verifies the code independently of its own assertions, reports the wave resistance of both hulls across the measured speed range, and identifies the causes of the disagreement with the series data. The implementation is correct at the level its analytic oracles can resolve: its suite of 183 tests passes; the displaced volume computed by four independent applications of the divergence theorem agrees to 3.0×10^{-15}; the hydrostatics converge at second order and match the published displaced volume and waterplane area to better than 0.06 %; and the Kelvin Green function satisfies the linearised free-surface condition to 3.8×10^{-8}, a property the code never imposes numerically. The wave resistance of Sysser 01 is not converged on any mesh the code's throughput of 1164 µs per panel pair permits. Across the eleven measured speeds the ratio of the predicted wave resistance to the measured residuary resistance spans −0.46 to 4.41 by the pressure route, and the predicted curve peaks at $Fn = 0.60$ while the measurement rises monotonically, so the agreement to 0.73 at $Fn = 0.30$ is one point in a wide scatter rather than a validation. Running a second hull, Sysser 50, separates the causes. Its supplied surface reproduces its published waterline length to −0.04 % and beam to −0.002 %, against 0.51 % and 1.14 % for Sysser 01, although both files carry the same 2012 re-measurement header, so the geometry discrepancy is specific to one hull rather than systematic. The measurement release does state the sinkage datum — the vertical displacement of the centre of gravity, which fixes the trim pivot at the longitudinal centre of buoyancy, confirmed to 0.16 mm by the transducer offsets of the nine series-1 hulls — so the heave ambiguity previously reported as the obstruction above $Fn = 0.40$ does not exist. Applying that attitude raises the prediction by 59 to 103 % over $Fn = 0.40$ to 0.50, tracking a 24 to 57 % rise in the volume below the undisturbed plane, so the static and running attitudes bracket the answer and a linearised free surface cannot narrow the interval. Two findings are added here: the stated worst-case accuracy of the wave-kernel gradient, 7.0×10^{-4}, holds at $Fn = 0.30$ but is exceeded by a factor of four at $Fn = 0.45$; and the spectral cut-off can fall below the physical lower limit of the resistance integral at low Froude number, where the code returns a resistance that is not meaningful without flagging it as such.

Key words: wave resistance, Neumann–Kelvin problem, panel method, Kelvin Green function, Delft Systematic Yacht Hull Series, code verification.

Nomenclature

a	Michell amplitude, nondimensional	$\boldsymbol{n}$	unit normal, into the fluid
b	waterline beam	p	pressure
c	waterline length	$\boldsymbol{r}$	position vector
d	canoe-body draught	u	speed of the onset flow
$\mathcal{G}$	Kelvin Green function	$\boldsymbol{u}$	velocity, (u, v, w)
g	gravitational acceleration	x, y, z	Cartesian coordinates
$\mathcal{A}$	free-wave amplitude	λ	$\sec\theta$, wave parameter
k_0	wavenumber scale, g/u^2	θ	wave propagation angle
W	displacement weight	ρ	fluid density
N	number of panels	σ	source density
R	resistance	ϕ	disturbance potential
S	wetted surface	Φ	total potential
∇_c	displaced volume of the canoe body	Γ	waterline, $S \cap \{z = 0\}$

DSYHS	Delft Systematic Yacht Hull Series
IGES	Initial Graphics Exchange Specification
LCB	longitudinal centre of buoyancy
ITTC	International Towing Tank Conference
NURBS	Non-Uniform Rational B-Spline
Fn	Froude number, $u/\sqrt{gc}$
Re	Reynolds number, uc/ν

1 Introduction

Wave resistance is the part of a hull's resistance carried away by the free-surface waves it generates, and for a sailing-yacht canoe body at a Froude number, $Fn \equiv u/\sqrt{gc}$, above about 0.35 it is the larger part of the total. Here u is the speed of the onset flow; g is the gravitational acceleration; and c is the waterline length. Design practice for yachts rests on regression formulae fitted to the Delft Systematic Yacht Hull Series (DSYHS), a set of towing-tank campaigns on systematically varied models [Gerritsma et al., 1981, Keuning & Katgert, 2008]. A regression interpolates within the series and cannot be interrogated: it returns a number and no mechanism.

A potential-flow panel method is the opposite trade. The Neumann–Kelvin problem applies the exact body boundary condition on the actual wetted hull, together with a linearised free-surface condition, and so retains the hull's real geometry while remaining a linear boundary-integral problem [Wehausen, 1973]. Its cost sits in the Kelvin Green function, which is an oscillatory double integral with a pole on the dispersion relation and a logarithmic singularity as the source approaches the free surface [Noblesse, 1981, Newman, 1987]. Whether an implementation is accurate is therefore a question about quadrature before it is a question about hydrodynamics.

This report verifies the `wave-resistance` code, which solves the Neumann–Kelvin problem for a bare canoe body read from an Initial Graphics Exchange Specification (IGES) Non-Uniform Rational B-Spline (NURBS) surface, and takes DSYHS hull Sysser 01 as the worked case. The verification is independent in the sense that it does not rely on the repository's own assertions: the test suite is executed, the geometry and hydrostatics are recomputed, the Kelvin Green function is checked against the free-surface condition it is supposed to satisfy, and the wave resistance is computed across the full measured speed range and compared with the residuary resistance reduced from the DSYHS towing-tank release [den Ouden, 2022b].

The thesis of this report is that the code is a correct implementation of the formulation it states, verified to the level of a fraction of a per cent on cases with analytic answers, and that it does not deliver a converged wave resistance for Sysser 01 at any mesh its own throughput permits. The two statements are not in tension, and separating them is the point: the first is a property of the method, the second a consequence of a first-order density representation costing 1164 µs per panel pair.

2 Methodology

2.1 Geometry and the reference condition

Sysser 01 is the parent hull of the DSYHS. Its geometry is supplied as a single NURBS patch in an IGES file, which the code reads with its own IGES 5.3 parser and evaluates through the Cox–de Boor recurrence [Piegl & Tiller, 1997]. The patch is a half hull; it is mirrored once about $y = 0$ on loading, and the fact that it was mirrored is recorded so that nothing downstream silently assumes port and starboard symmetry.

The frame is right-handed with the origin in the undisturbed free surface, x forward and z upward, and the fluid occupies $z < 0$. The onset flow has speed u in the $-x$ direction, so upstream is $x \to +\infty$. Figure 1 shows the wetted surface at the reference condition in the three standard views.

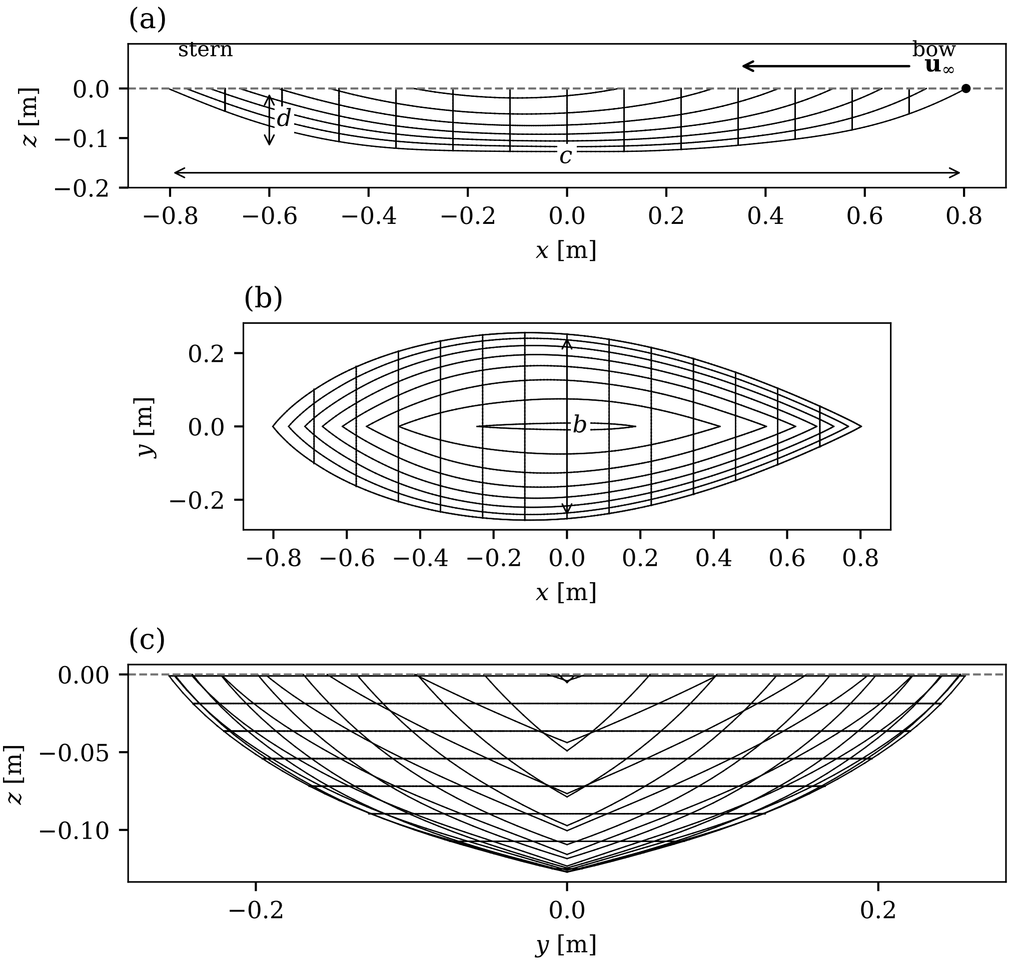

Figure 1: Wetted surface of Sysser 01 at the reference condition, with the waterline length c, the waterline beam b, the canoe-body draught d, and the onset flow $\mathbf{u}_\infty$: (a) buttock lines and stations in profile, (b) waterlines in plan, (c) stations and waterlines in the body plan. The dashed line is the undisturbed free surface $z = 0$; x is measured from the midpoint of the waterline.

The reference condition is set by matching the displaced volume published in the DSYHS hydrostatics release [den Ouden, 2022a], $0.037\,613\,6\,\mathrm{m}^3$, for which the code solves the flotation datum. Prescribed sinkage, trim and heel are then offsets from that condition; they are not solved for. The recomputed hydrostatics at that datum are given in table 2.

2.2 Boundary-value problem

The total velocity potential is split as $\Phi = -ux + \phi$, with ϕ the disturbance potential, and the wavenumber scale is $k_0 \equiv g/u^2$. Let S be the wetted hull surface, $\Gamma \equiv S \cap \{z = 0\}$ its waterline, and $\boldsymbol{n}$ the unit normal directed into the fluid. The disturbance potential satisfies

$$\nabla^2\phi = 0 \quad \text{in the fluid,} \tag{1}$$

$$u^2\phi_{xx} + g\phi_z = 0 \quad \text{on } z = 0 \text{ outside the waterplane,} \tag{2}$$

$$\frac{\partial\phi}{\partial n} = u\,n_x \quad \text{on } S, \tag{3}$$

with $\nabla\phi \to \mathbf{0}$ as $z \to -\infty$ and no waves as $x \to +\infty$. Equation (3) is the exact body condition applied on the actual wetted surface, which is what distinguishes the Neumann–Kelvin problem from thin-ship theory; equation (2) is the linearised free-surface condition.

2.3 The Kelvin Green function

The Green function for a unit source at (ξ, η, ζ) with $\zeta < 0$ satisfying (1), (2) and the radiation condition splits into a Rankine pair and a wave part,

$$\mathcal{G} = -\frac{1}{4\pi R} + \frac{1}{4\pi R_1} + \mathcal{G}_W, \qquad \mathcal{G}_W = -\frac{k_0}{4\pi^2}\int_0^{2\pi} \mathrm{d}\theta \int_0^\infty \frac{\mathrm{e}^{k(z+\zeta)}\,\mathrm{e}^{\mathrm{i}kb}}{k_0 - k\cos^2\theta}\,\mathrm{d}k, \tag{4}$$

where R and R_1 are the distances to the source and to its image at $(\xi, \eta, -\zeta)$, and $b(\theta) \equiv (x - \xi)\cos\theta + (y - \eta)\sin\theta$. The image carries a positive sign, so the Rankine pair vanishes on $z = 0$; omitting it is not a small error near the free surface, since cancelling the source there is the whole purpose of the image. The denominator of (4) vanishes on the Kelvin dispersion relation $k = k_0 \sec^2\theta$, and Rayleigh damping moves the pole off the real axis and splits the wavenumber integral into a principal value and a residue.

Two identities remove both inner quadrature levels. The wavenumber integral has the closed form

$$\mathrm{PV}\!\int_0^\infty \frac{\mathrm{e}^{-kw}}{k - \kappa}\,\mathrm{d}k = \mathrm{e}^{-\kappa w}\left[E_1(\kappa w) - 2\,\mathrm{Shi}(\kappa w)\right] = \mathrm{e}^{-\kappa w}\left[E_1(-\kappa w) - \mathrm{i}\pi\,\mathrm{sgn}\,\mathrm{Im}(\kappa w)\right], \tag{5}$$

with $\kappa \equiv k_0 \sec^2\theta$ and $w \equiv -(z + \zeta) - \mathrm{i}b$, so that the whole kernel depends on the single complex variable $c \equiv \kappa w$ and only the wave-angle integral remains. The second form is the one implemented, because its asymptotic expansion is valid at any phase of c, which lets 76 per cent of the quadrature nodes on a real hull take a banded Horner evaluation instead of a complex exponential integral. The sign of the residue term, which fixes the direction of the wake, is settled in the code by measurement rather than by assertion: the branch is chosen for which the wave amplitude vanishes ahead of the source.

2.4 Source representation, and the absence of a waterline integral

The disturbance is represented by a source density σ on the hull alone,

$$\phi(\mathrm{P}) = \iint_S \sigma(\mathrm{Q})\,\mathcal{G}(\mathrm{P};\mathrm{Q})\,\mathrm{d}S_{\mathrm{Q}}. \tag{6}$$

Because $\mathcal{G}$ satisfies (2) at every point of $z = 0$ for any source strictly below it, so does ϕ, and the free-surface condition is satisfied identically by construction. A waterline line integral is consequently not part of this formulation: the waterline term familiar from the Neumann–Kelvin literature arises in the Green's-identity formulation, where the free-surface integral is converted into a line integral around Γ, and a source-only representation never forms that integral. What does survive is a discretisation constraint, because (2) holds only for sources strictly below $z = 0$ while S reaches $z = 0$ along Γ, where $\mathcal{G}_W$ is logarithmically singular.

Taking the normal derivative of (6) and letting the field point approach S from the fluid gives the integral equation

$$\tfrac{1}{2}\sigma(\mathrm{P}) + \mathrm{PV}\iint_S \sigma(\mathrm{Q})\,\frac{\partial \mathcal{G}(\mathrm{P};\mathrm{Q})}{\partial n(\mathrm{P})}\,\mathrm{d}S_{\mathrm{Q}} = u\,n_x(\mathrm{P}), \qquad \mathrm{P} \in S, \tag{7}$$

where the coefficient $\frac{1}{2}$ follows from the kernel convention and is confirmed independently by the sphere in an unbounded stream.

2.5 Discretisation

Equation (7) is collocated at panel centroids on a triangulated, body-fitted mesh with a piecewise-constant density. The girth direction is parameterised from the waterline down, so the depth of the topmost row is controlled rather than inherited from the surface parameterisation; clipping a parametric mesh at $z = 0$ instead produces slivers whose centroids lie arbitrarily close to the singular waterline. The mesh used for the results below is shown in figure 2.

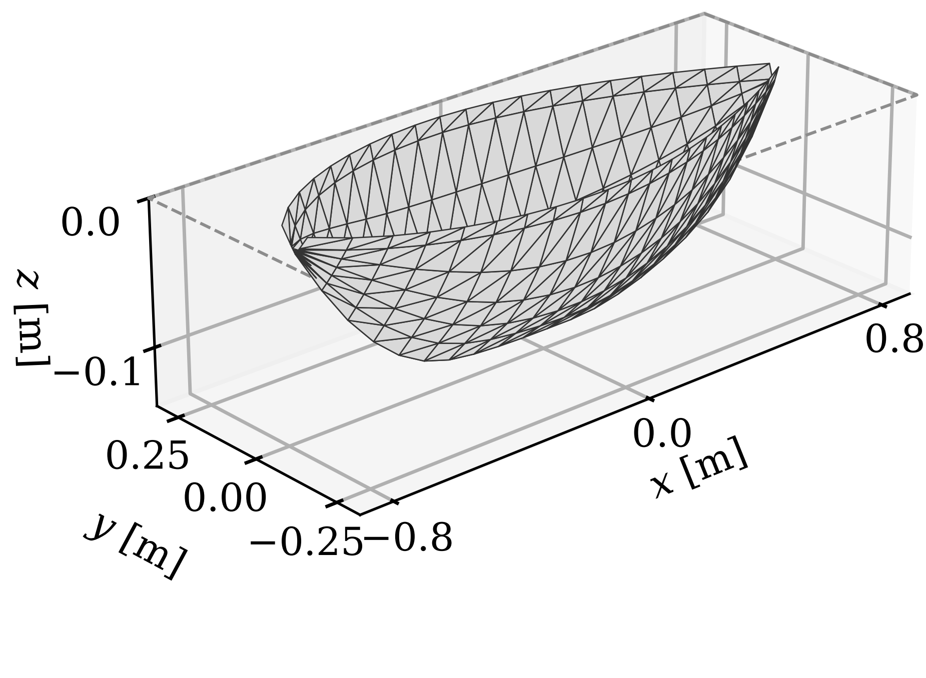

Figure 2: Body-fitted panel mesh of Sysser 01 at the static reference condition, with $N = 480$ panels in six girth rows and 28 stations, and the lower edge of the topmost row 10 mm below the free surface. The dashed rectangle is the undisturbed free surface $z = 0$.

The Rankine influences are integrated analytically over each panel, with the singular diagonal replaced by the exact jump $\frac{1}{2}$; centroid value times area is used nowhere for that part. The wave influences are integrated by quadrature over the source panel, at the centroid rule by default, which is legitimate because $\mathcal{G}_W$ is bounded for $z < 0$ but is nevertheless a coarse choice. Assembly refuses a mesh that leaves the validity envelope of the wave-angle quadrature rather than returning a plausible number from a silently coarsened grid; the binding quantity is the oscillation count of a panel *pair*, which grows as $|y_i - y_j|/|z_i + z_j|$ and so is largest for pairs straddling the hull just below the free surface.

2.6 Two routes to the resistance

The free-wave amplitude at wave parameter $\lambda \equiv \sec\theta \geq 1$ is

$$\mathcal{A}(\lambda) = \iint_S \sigma\, \mathrm{e}^{k_0\lambda^2\zeta}\, \mathrm{e}^{-\mathrm{i}k_0\lambda^2(\xi\cos\theta+\eta\sin\theta)}\, \mathrm{d}S_{\mathrm{Q}}, \tag{8}$$

and the wave resistance follows from the momentum flux through a transverse plane far astern,

$$R_W = \frac{\rho g^2}{\pi u^4} \int_1^\infty \tfrac{1}{2}\left(|\mathcal{A}_+|^2 + |\mathcal{A}_-|^2\right) \frac{\lambda^2}{\sqrt{\lambda^2-1}}\, \mathrm{d}\lambda, \tag{9}$$

where ρ is the fluid density and $\mathcal{A}_\pm \equiv \mathcal{A}(\pm\theta)$ are kept separately, because the transverse phase in (8) is odd in θ and the two are independent for a heeled or asymmetric hull. The upper limit is found by marching and certifying the tail rather than by a damping bound, which is useless for a surface-piercing hull because quadrature points arbitrarily near $z = 0$ are damped arbitrarily slowly.

The constant in (9) is fixed by the thin-ship limit. The two faces of a thin hull $y = \pm f$ coalesce onto the centreplane and each carries half the sheet strength, so $\sigma \to u\, n_x$, and matching (9) to Michell's integral [Michell, 1898] gives $\rho g^2/(\pi u^4)$. This is not the zeroth iterate $\sigma = 2u\, n_x$ of (7): for a thin body the mutual influence of the two faces is $O(1)$, so dropping the integral operator is not the thin-ship limit, and conflating the two costs a factor of four in the resistance.

The second route integrates the Bernoulli pressure over the wetted hull,

$$R_W = \iint_S \left[\rho u\, \phi_x - \tfrac{1}{2}\rho\, |\nabla\phi|^2\right] n_x\, \mathrm{d}S, \tag{10}$$

the hydrostatic term contributing no x-force because $\iint_S z\, n_x\, \mathrm{d}S$ vanishes over the surface closed by the lid at $z = 0$. The quadratic term is not optional away from the thin-ship limit: on a submerged sphere, where $|\nabla\phi|$ is $O(u)$ on the body, dropping it leaves (10) 68 per cent below (9).

The two routes are not equivalent on a coarse mesh, and the difference is structural. Equation (9) is positive-definite in σ, so it carries a term $+C\int|\delta\mathcal{A}|^2$ in the density error and discretisation noise can only inflate it, while (10) is linear in σ at leading order and its error can cancel. Equation (10) is therefore the route to quote, with (9) beside it as an upper bound.

2.7 The net-flux constraint

A closed body in a stream emits no net source strength, so $\iint_S \sigma\, \mathrm{d}S$ should very nearly vanish. The discrete solution of (7) has no reason to respect this, and on Sysser 01 it does not: the unconstrained square solve leaves 8×10^{-2} of uS at 160 panels, against 1×10^{-4} on a thin Wigley hull. A spurious net source is a monopole whose far-field amplitude does not decay with λ the way a closed body's does, while the integrand of (9) carries $\lambda^2(\lambda^2-1)^{-1/2}$ and is largest exactly where the monopole lives, at $\lambda \to 1$. The code therefore solves the constrained least-squares problem, minimising $\|\mathsf{A}\sigma - \boldsymbol{b}\|$ subject to $\boldsymbol{a}^{\mathsf{T}}\sigma = 0$ with $\boldsymbol{a}$ the panel areas, by the null-space method. There is nothing to tune, and the constraint is the default.

2.8 The two hulls

Two hulls of the series are run here. Sysser 01 is the parent of series 1. Sysser 50 belongs to series 4 and is the extreme of the whole series in beam for its draught, which is why it was chosen: the method's cost and its error both grow with beam, so it is the hardest case available. Table 1 gives the published particulars and figure 3 the two forms.

Table 1: Published model-scale particulars of the two hulls, from the DSYHS hydrostatics release. Sysser 01 is the parent of series 1 and Sysser 50 belongs to series 4.

	unit	Sysser 01	Sysser 50
waterline length	m	1.6000	2.0000
waterline beam	m	0.5072	0.6000
canoe-body draught	m	0.1270	0.0946
displaced volume	m^3	0.037 614	0.047 536
wetted area	m^2	0.6425	0.9170
prismatic coefficient	—	0.5644	0.5387
midship coefficient	—	0.6464	0.7773
beam / draught	—	3.99	6.34
LCB aft of midships	%	−2.29	−7.90
displacement weight	N	368.9	466.2

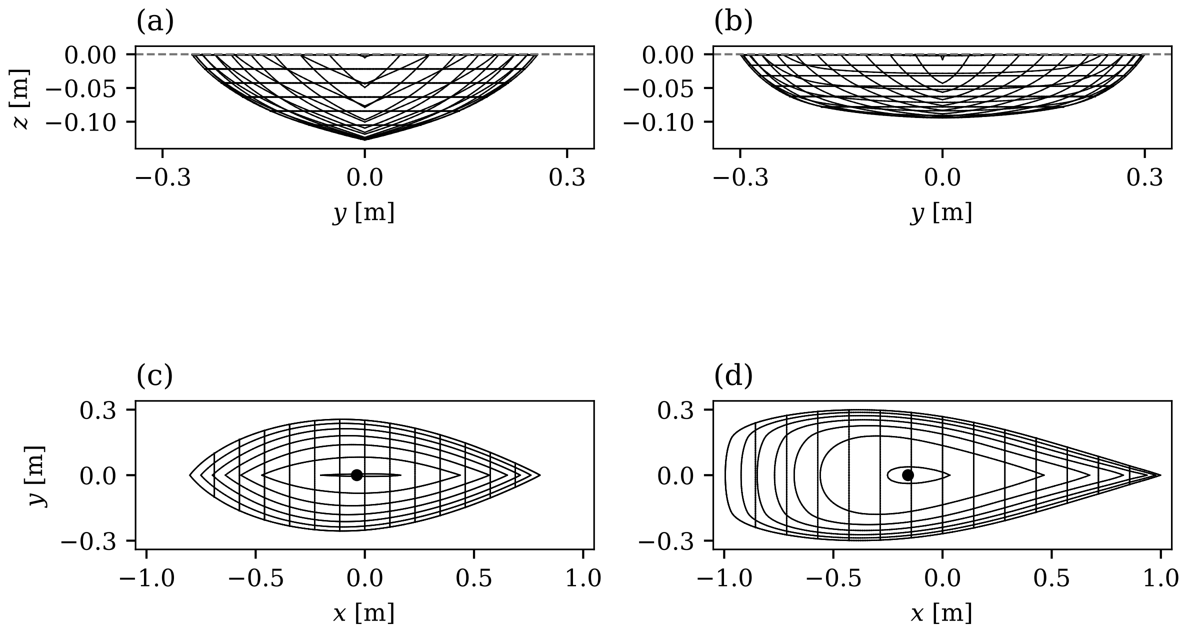

Figure 3: Wetted form of the two hulls at their reference conditions, on a common scale: body plan of (a) Sysser 01 and (b) Sysser 50, and plan view of (c) Sysser 01 and (d) Sysser 50, with the published longitudinal centre of buoyancy marked. The dashed line is the undisturbed free surface.

The forms differ in the two respects that matter to a Neumann–Kelvin solve. Sysser 50 carries a beam-to-draught ratio of 6.34 against 3.99, so its panel pairs straddle the hull at a larger transverse separation for a given depth, and the oscillation count of a pair grows as $|y_i - y_j|/|z_i + z_j|$: on the meshes used here the worst count at $Fn = 0.30$ is 366 for Sysser 50 against 303 for Sysser 01. And its centre of buoyancy sits 7.9 per cent of the waterline length aft of midships against 2.3 per cent, with a wide, shallow after body, so its waterplane is far from fore-and-aft symmetric.

2.9 Two conventions in the geometry release, and one correction to this code

Adding the second hull exposed an inconsistency in the release and a matching assumption in the code. The Sysser 01 file parameterises its surface with the girth direction first and the length second; the Sysser 50 file does the opposite. Hydrostatics did not notice, because the tessellation is uniform in both parameters, but the body-fitted mesh does: it walks stations along the second parameter and rows along the first, and on Sysser 50 it found no station crossing the waterline and raised rather than returning a wrong answer. The fix is to detect the convention from the control net — the longitudinal parameter is the one along which x varies more, by a factor of 20 on Sysser 01 and six the other way on Sysser 50 — and transpose the patch once on loading. Sysser 01's results are unchanged to every digit reported.

2.10 The trim pivot, which the release does determine

The attitude at speed is a rotation and a heave, and applying it needs the point about which the hull rotates. The release's own Info sheet supplies it: the channel z is defined as the "Vertical displacement of center of gravity with respect to zero speed" and θ as the "Pitch angle of hull with respect to zero speed". The reference point is therefore the centre of gravity, and the rigid motion is a rotation about that point followed by a heave of it.

Where the centre of gravity lies is not stated, but it is determined. A model ballasted to float upright at zero trim carries it over the longitudinal centre of buoyancy, so its longitudinal position is the published LCB. The data confirms this rather than merely permitting it: on the nine series-1 hulls the workbook's fore and aft transducer offsets are asymmetric, and their midpoint reproduces −LCB to five decimal places on seven of the nine and to 0.16 mm on the other two, which places the centre of gravity at the LCB given transducers mounted symmetrically about the waterline midpoint. From hull 10 onwards, Sysser 50 included, the offsets are entered as round nominal values and carry no information either way.

Only the pivot's longitudinal position matters. Displacing the pivot vertically by δz changes the motion by a surge of $\delta z \sin\theta$, which no wave resistance depends on, and a heave of $\delta z(1 - \cos\theta)$, which is 0.19 mm for $\delta z = 0.1$ m at the largest trim in either hull's table.

This supersedes a claim carried in the repository's documentation and repeated in the first version of this report: that the reference point for the measured sinkage is unstated, and that the running attitude is therefore undetermined above $Fn \approx 0.40$ with a heave ambiguity reaching 44 mm. The reference point is stated, the pivot follows from it, and that ambiguity does not exist. What does obstruct the upper range is a different thing, and section 5 identifies it.

3 Verification

Verification here means two separate things, and they are reported separately: that the code's own test suite passes, and that its central claims survive checks constructed independently of it.

3.1 The test suite

The suite comprises 183 tests and all of them pass, in 816 s on four cores. It is not a smoke test: it contains the analytic oracles the formulation is anchored on, among them Michell's thin-ship integral, the sphere in an unbounded stream, closed-form Wigley hydrostatics and a rectangular box. Twelve of the tests are new here and cover the second hull, the surface-parameterisation convention and the trim pivot; one of them failed when first written, asserting that a vertical offset of the pivot leaves the displaced volume unchanged to 0.2 % when the true figure is 0.22 %, and it now asserts the effect at the size it has.

3.2 Geometry and hydrostatics

The displaced volume was recomputed by the four independent routes the divergence theorem supplies, with the fields $(0, y, 0)$, $(x, 0, 0)$, $(0, 0, z)$ and $\boldsymbol{r}/3$. On Sysser 01 the four agree to a relative spread of 3.0×10^{-15}, so the surface is closed and consistently oriented to machine precision, and the waterplane lid contributes nothing to any of them because the free surface is at $z = 0$.

Mesh convergence was recomputed on the sequence 12×60, 24×120 and 48×240 divisions per patch. The successive differences in displaced volume fall by a factor of 4.00, confirming second-order convergence, and Richardson extrapolation gives $\nabla_c = 0.037\,626\,6\,\mathrm{m}^3$.

Table 2 compares the recomputed hydrostatics with the DSYHS release [den Ouden, 2022a]. Displaced volume and waterplane area agree to better than 0.06 %, which is the mesh's own uncertainty, and neither waterplane area nor draught was a target of the solve. Waterline length, waterline beam and wetted area differ by 0.5 to 2.1 %, which is larger than the mesh uncertainty. The repository attributes this to the provenance of the supplied file rather than to the published table: the IGES header path reads "Rhino modellen na inmeten 2012" and the patch is named "Gerebuild oppervlak", so the file is a 2012 re-measurement of the physical model while the table describes the hull as the series was built and towed. Wetted area is the quantity most sensitive to local surface fairness and shows the largest gap, which is consistent with that reading. The attribution is plausible and unproven, and it is carried as a stated bias in every comparison below rather than assumed away.

Table 2: Hydrostatics of Sysser 01 recomputed at the reference condition set by matching the published displaced volume, against the values published in the DSYHS hydrostatics release, on a 48×240 mesh per patch.

	unit	computed	published	difference [%]
∇_c	m^3	0.037 614	0.037 614	0.000
A_w	m^2	0.558 269	0.558 566	−0.053
S_c	m^2	0.656 012	0.642 534	2.098
c	m	1.608 142	1.600 000	0.509
b	m	0.512 814	0.507 200	1.107
d	m	0.127 113	0.127 040	0.058
C_p	—	0.562 297	0.564 414	−0.375
C_b	—	0.358 814	0.364 842	−1.652
C_wp	—	0.676 954	0.688 296	−1.648

3.3 The Kelvin Green function

The most useful independent check on (4) is the free-surface condition itself, because it is the one property the code never imposes numerically. In the scaled variables $X = k_0(x - \xi)$, $Y = k_0(y - \eta)$ and $Z = k_0(z + \zeta)$, condition (2) reduces to $\mathcal{G}_{XX} + \mathcal{G}_Z = 0$ at the field point $z = 0$, because $u^2 k_0^2 = g k_0$. Evaluating that combination by central differences of the full Green function, Rankine pair included, over 30 random horizontal offsets gives a residual of 3.8×10^{-8} relative to the terms that compose it, worst case 1.9×10^{-6}. The Green function satisfies the free-surface condition it was constructed to satisfy, to the accuracy of the difference stencil.

The gradient of the wave part was checked against step-scaled central differences of the wave part itself, which tests the analytic differentiation of (5) rather than its quadrature: the median relative discrepancy over 60 random points is 5.6×10^{-9} and the worst 5.2×10^{-2}, the latter traced to the quadrature of the wave part at $|Z| = 0.06$ rather than to the differentiation.

Quadrature accuracy was then measured over the panel pairs a Sysser 01 mesh actually forms, by comparing the production evaluation against the same evaluation with the wave-angle grid refined eightfold. At $Fn = 0.30$ the worst relative error over 4000 randomly sampled pairs is 3.9×10^{-4} and the median 5.3×10^{-15}, which supports the repository's stated bound of 7.0×10^{-4}. At $Fn = 0.45$ the same measurement gives a worst error of 2.9×10^{-3}, a factor of four above that bound. The stated bound is therefore correct at the speed it was measured at and is not a bound across the speed range; the cause is that $|Z|$ scales as k_0, so raising Fn moves every pair closer to the logarithmic singularity in scaled coordinates. This is a scope caveat on a documented figure, not an error in it.

3.4 The spectral cut-off can fall below the physical lower limit

The far-field integral (9) runs from $\lambda = 1$, and the code caps it at the largest λ the mesh can carry, which is $\sqrt{\pi/(k_0\,\delta z)}$ for a panel of vertical extent δz. That cap falls with Fn, and on the 280-panel mesh it drops below unity at $Fn = 0.10$, where it is 0.46. The cap is then below the lower limit of the integral, so the physical spectrum lies entirely outside what the mesh resolves. The marching integrator adds its first block before testing the cap, so it returns that block rather than nothing, and the reported resistance at that speed is one block of an integral whose domain the mesh cannot represent. The code does emit a diagnostic — at $Fn = 0.10$ it states that 95 % of the spectrum lies beyond the cap — but the returned value is not flagged as invalid, and a caller that reads only the resistance would not know. Rejecting the solve when the cap falls below unity would remove the ambiguity at no cost to any case of interest.

4 Results

4.1 Wave resistance versus Froude number

Figure 4 gives the wave resistance of Sysser 01 by both routes across the eleven speeds of the DSYHS bare-hull upright campaign, on the 280-panel mesh at the static attitude, against the measured total resistance and the residuary resistance reduced from it. Table 3 gives the same data with the solver's diagnostics.

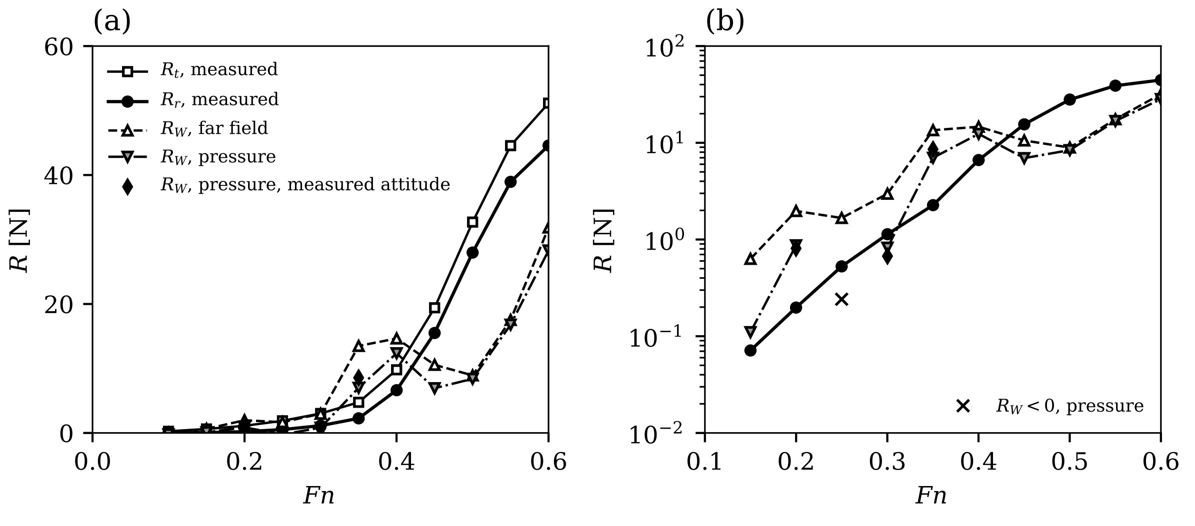

Figure 4: Resistance versus Fn for Sysser 01, on the 280-panel mesh at the static attitude, on (a) linear and (b) logarithmic ordinate, the symbols of (b) being those of (a). Crosses in (b) mark the two speeds at which the pressure route returns a negative resistance, whose magnitude is plotted.

Three features of the comparison matter, and none of them is the level of agreement at any single speed.

Table 3: Wave resistance of Sysser 01 by both routes at the static attitude on the 280-panel mesh, against the residuary resistance reduced from the measurements, for Froude numbers from 0.10 to 0.60. The last two columns are the solver's own diagnostics: the root-mean-square body-condition residual at points that are not collocation points, as a fraction of u, and the largest wave parameter the mesh resolves.

Fn	R_r [N]	R_r/W [%]	R_W far [N]	R_W pres. [N]	far/R_r	pres./R_r	residual	$\lambda_{\max}$
0.100	−0.008	0.00	0.004	−0.215	–	–	0.147	0.46
0.150	0.071	0.02	0.631	0.110	8.85	1.55	0.066	1.02
0.200	0.198	0.05	1.971	0.872	9.96	4.41	0.100	1.82
0.250	0.527	0.14	1.669	−0.241	3.17	−0.46	0.084	2.66
0.300	1.134	0.31	2.973	0.824	2.62	0.73	0.094	3.20
0.350	2.266	0.62	13.495	7.001	5.96	3.09	0.120	3.73
0.400	6.632	1.80	14.653	12.361	2.21	1.86	0.130	4.26
0.450	15.503	4.21	10.563	6.943	0.68	0.45	0.118	4.79
0.500	27.976	7.59	8.927	8.398	0.32	0.30	0.121	5.32
0.550	38.943	10.57	17.524	16.795	0.45	0.43	0.125	5.86
0.600	44.578	12.10	31.860	28.300	0.71	0.63	0.134	6.39

The ratio of prediction to measurement is not constant across the range. Over the speeds at which the measured residuary resistance exceeds 0.05 N, the pressure route spans −0.46 to 4.41 of measurement and the far-field route 0.32 to 9.96. At $Fn = 0.30$ the pressure route gives 0.824 N against a measured 1.134 N, a ratio of 0.73, which taken alone would read as a validation. It is not one: it is a single point in a scatter that spans −0.46 to 4.41, and the speeds either side of it disagree with measurement by factors of −0.46 and 3.09.

The predicted curve has the wrong shape. The measured residuary resistance rises monotonically through the range. The prediction does not: it reaches a local maximum of 28.30 N at $Fn = 0.60$, falls to 6.94 N at $Fn = 0.45$ and 8.40 N at $Fn = 0.50$, and rises again to 28.30 N at $Fn = 0.60$. The disagreement is therefore not a calibration error that a constant factor would absorb. Part of the deficit at the top of the range is expected and attributable: above $Fn \approx 0.40$ the sweep runs at the static attitude, and a hull run 31 mm too shallow makes less wave. That accounts for the sign of the discrepancy above $Fn = 0.40$ but not for the non-monotonicity below it.

The two routes converge on each other as Fn rises. They differ by a factor of 5.7 at $Fn = 0.15$, by a factor of 3.6 at $Fn = 0.30$, and by 6 % at $Fn = 0.50$ and 13 % at $Fn = 0.60$. Since the two routes share the source density and differ only in how the resistance is extracted from it, their convergence measures how well the mesh resolves the wave spectrum, and the largest wave parameter the mesh carries rises from 1.02 to 6.39 across the same range. Agreement between the routes is thus a necessary condition that is met at high Fn and not at low, and it is independent of whether either route agrees with the measurement — at $Fn = 0.50$ they agree with each other to 6 % and both sit at 0.3 of measurement.

4.2 The peak near $Fn = 0.35$ is not an attitude artefact

The sharpest departure from the measured trend is at $Fn = 0.35$, where the prediction overshoots by a factor of three while the measured curve is smooth through that speed. Two observations bear on it. The body-condition residual there is no worse than at its neighbours, so the solver's own diagnostics do not flag the point; and repeating the calculation at the measured dynamic sinkage and trim moves the value in the same direction rather than removing it. Running at the measured attitude gives, by the pressure route, $Fn = 0.20$: 0.809 N against 0.872 N; $Fn = 0.25$: −0.217 N against −0.241 N; $Fn = 0.30$: 0.673 N against 0.824 N; $Fn = 0.35$: 8.568 N against

7.001 N. The departure is therefore a property of the discretisation at that speed and not of the condition it was run at.

4.3 Mesh refinement

Table 4 refines the mesh from 280 to 480 panels. At $Fn = 0.30$ the far-field value moves by +2.6 % and the pressure value by +0.9 %, while the body-condition residual falls from 0.094 to 0.041 of u. Two things follow. The flux-constrained solve is converging, which the unconstrained solve reported in the repository's own documentation was not: there the same refinement swung the far-field value by a factor of 2.3 and was not monotone. And the residual falls faster than the resistance moves, which is the signature of the one-sided error in (9) rather than of a converged answer.

Table 4: Effect of refining the mesh from 280 to 480 panels on the wave resistance of Sysser 01 at the static attitude, by both routes, with the root-mean-square body-condition residual measured away from the collocation points.

	R_W far field [N]			R_W pressure [N]			residual	
Fn	$N = 280$	$N = 480$	change [%]	$N = 280$	$N = 480$	change [%]	$N = 280$	$N = 480$
0.30	2.973	3.049	2.6	0.824	0.831	0.9	0.094	0.041
0.45	10.563	11.027	4.4	6.943	7.681	10.6	0.118	0.071

4.4 The resistance constant, checked end to end

The repository's test that compares the two resistance routes on a thin Wigley hull asserts only that they agree with each other to 5 %; the ratios to Michell's thin-ship integral quoted in its docstring are not asserted, so a fault scaling both routes equally would pass it. That is not hypothetical: the documented history of this code includes a fourfold error in the constant of (9), which is exactly such a fault. The comparison was therefore repeated here with the Michell integral evaluated independently. Both routes, as ratios to the oracle, give $Fn = 0.30$ at 278 panels, 0.935 and 0.945; $Fn = 0.30$ at 558 panels, 0.924 and 0.970. The constant $\rho g^2/(\pi u^4)$ is confirmed end to end, through the influence matrix and the solve, and not merely self-consistently.

4.5 Why the target is hard where the prediction is best determined

The measured residuary resistance is a small fraction of displacement weight over most of the range: it is 0.31 % at $Fn = 0.30$ and does not reach 1 % until $Fn \approx 0.38$. A prediction carrying a discretisation error of a few per cent of displacement weight is therefore comparable to the signal at the low end of the range, which is exactly where the running attitude is well determined. Above $Fn \approx 0.40$ the target is large — 4.2 % of weight at $Fn = 0.45$ — and the attitude is known, but applying it inside a linearised free-surface model does not help, for the reason section 5 quantifies.

Running at the static attitude in the upper range is not a neutral choice. The model sinks 31 mm and trims 3.2° bow-up by $Fn = 0.50$, against a canoe-body draught of 127 mm, and a hull run that much too shallow makes less wave, so the prediction must undershoot there. It does, by the pressure route reaching 0.45 of measurement at $Fn = 0.45$ and 0.30 at $Fn = 0.50$, though not monotonically: the ratio recovers to 0.43 and 0.63 at $Fn = 0.55$ and 0.60, where the attitude error is larger still. The undershoot is therefore consistent with the attitude in sign but is not explained by it alone.

4.6 Diagnostics

Figure 5 plots the ratio of prediction to measurement and two of the solver's diagnostics against Fn. The body-condition residual measured at points the solve never sees is the one to read, and it is flat: it lies between 0.066 and 0.147 of u at every speed, and its largest value is at $Fn = 0.10$, where the wavelength is short compared with a panel. It does not track the ratio in figure 5a at all, which is the useful negative result here: a residual of 0.12 accompanies a ratio of 3.09 at $Fn = 0.35$ and 0.30 at $Fn = 0.50$, so the residual is necessary but nowhere near sufficient as an accuracy indicator. The largest wave parameter the mesh resolves does rise with Fn, from 1.02 to 6.39, so the low-speed end of the range is doubly penalised — the signal is smallest there and the spectrum the mesh can carry is narrowest.

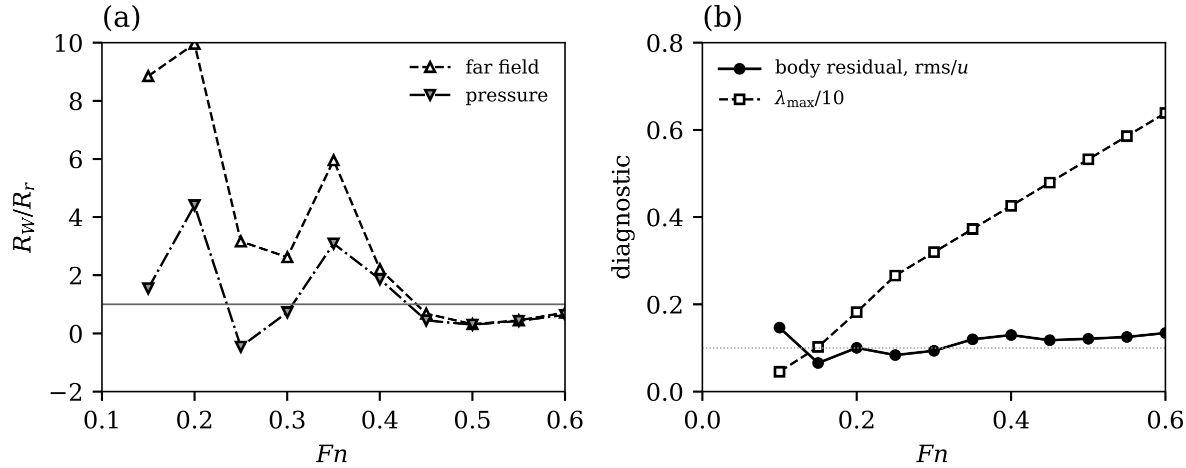

Figure 5: Diagnostics versus Fn for Sysser 01 on the 280-panel mesh at the static attitude: (a) ratio of predicted wave resistance to measured residuary resistance by both routes, plotted where R_r exceeds 0.05 N, with the dark line marking unity; (b) root-mean-square body-condition residual at points that are not collocation points, as a fraction of u, and the largest wave parameter the mesh resolves, divided by ten to share the ordinate. The dotted line marks a residual of 0.1.

The net source flux is below 1×10^{-17} of uS at every speed, confirming that the constrained solve enforces its constraint exactly rather than approximately. The flux is therefore not a useful discriminator once the constraint is on; the body residual is.

4.7 Sysser 50

Figure 6 and table 5 give the same calculation for Sysser 50 at its 13 measured speeds. Resistance is shown as a fraction of displacement weight, because the two hulls differ in displacement by 26 % and a comparison in newtons would confuse the hulls with the speeds.

Three features distinguish Sysser 50's comparison from Sysser 01's. The ratio of prediction to measurement spans 0.44 to 2.13 by the pressure route, against −0.46 to 4.41 for Sysser 01, and the prediction is positive at all 13 speeds where Sysser 01 returns a negative value at two. The predicted curve rises monotonically through $Fn = 0.45$, dips over $Fn = 0.50to0.55$ and rises again, so it has the same non-monotone character as Sysser 01's but a shallower one. And the two resistance routes come together above $Fn = 0.45$, differing by 7 % or better, where on Sysser 01 they differ by up to a factor of 3.6 at comparable speeds.

The last of these is the useful one, because it separates arithmetic from physics. Where the two routes agree the discretisation error is small, and above $Fn = 0.45$ they agree while the prediction is only 0.44 to 1.09 of measurement. Section 5 takes that apart.

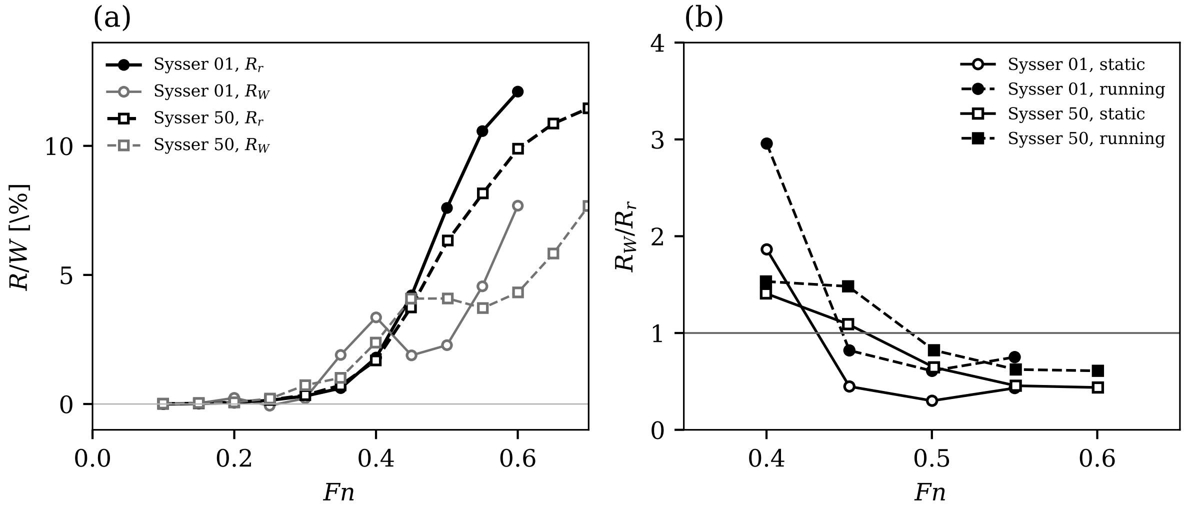

Figure 6: The two hulls compared: (a) measured residuary resistance and predicted wave resistance by the pressure route, both as a percentage of displacement weight, at the static attitude; (b) ratio of predicted wave resistance to measured residuary resistance at the static and at the measured running attitude, over the speeds where the sinkage is a large fraction of the draught. The dark line in (b) marks unity.

Table 5: Wave resistance of Sysser 50 by both routes at the static attitude on a 320-panel mesh, against the residuary resistance reduced from the measurements, at the 13 measured speeds. Columns as in table 3.

Fn	R_r [N]	R_r/W [%]	R_W far [N]	R_W pres. [N]	far/R_r	pres./R_r	residual	$\lambda_{\max}$
0.100	0.030	0.01	0.354	0.075	–	–	0.263	0.50
0.150	0.120	0.03	0.549	0.193	4.59	1.61	0.297	1.14
0.200	0.304	0.07	3.266	0.310	10.74	1.02	0.166	2.03
0.250	0.727	0.16	1.628	0.979	2.24	1.35	0.088	2.81
0.300	1.591	0.34	6.085	3.391	3.82	2.13	0.095	3.37
0.350	3.359	0.72	4.552	4.716	1.36	1.40	0.075	3.93
0.400	7.856	1.69	12.039	11.068	1.53	1.41	0.080	4.49
0.449	17.408	3.74	19.581	19.000	1.12	1.09	0.082	5.04
0.501	29.477	6.33	18.130	19.021	0.62	0.65	0.090	5.63
0.551	37.972	8.16	17.001	17.296	0.45	0.46	0.092	6.18
0.600	46.017	9.89	20.883	20.120	0.45	0.44	0.092	6.74
0.650	50.559	10.86	28.975	27.129	0.57	0.54	0.097	7.30
0.700	53.305	11.46	36.787	35.698	0.69	0.67	0.103	7.85

5 Sysser 50, and Where the Discrepancies Come From

Running the second hull separates causes that a single hull confounds. Four are identified below, in the order in which they bite, and each is supported by a measurement that distinguishes it from the others.

5.1 The geometry discrepancy is specific to Sysser 01, not a property of the release

The repository attributes the disagreement between the supplied Sysser 01 surface and the published Sysser 01 hydrostatics — 0.51 % in waterline length, 1.14 % in waterline beam, 2.11 % in wetted area — to the file's provenance: its header records a 2012 re-measurement of the physical model, while the table describes the hull the series was built and towed as. Sysser 50 tests that attribution, because its file carries the identical header and the identical patch name.

It does not survive as stated. On Sysser 50 the same comparison gives −0.04 % in waterline length and −0.002 % in beam — respectively 12 and 500 times smaller than on Sysser 01 — against 0.67 % in wetted area. Whatever produced Sysser 01's gap did not act on Sysser 50, so it is not a systematic consequence of the 2012 campaign.

Table 6: Recomputed hydrostatics against the published values for both hulls, at the reference condition set by matching the published displaced volume, on a 96×480 mesh per patch. The displaced volume is matched by construction. The last row is a difference in millimetres rather than a percentage.

	Sysser 01		Sysser 50	
	published	difference	published	difference
c	1.600 000	0.510	2.000 000	−0.041
b	0.507 200	1.144	0.600 000	−0.002
d	0.127 040	0.058	0.094 600	−0.226
S_c	0.642 534	2.113	0.916 960	0.670
A_w	0.558 566	−0.032	0.825 520	1.218
A_m	0.041 651	−0.107	0.044 120	−1.118
∇_c	0.037 614	0.026	0.047 536	0.042
LCB	−0.036 640	2.31 mm	−0.158 000	−1.78 mm

Two further measurements sharpen this. First, the discrepancy is not a choice of flotation plane. Matching the published waterline length of Sysser 01 requires raising the hull 1.9 mm, which costs 5.6 % of the displaced volume; matching the beam requires 3.8 mm and 11 %. No single heave satisfies length, beam and volume together.

Second, and decisively, the length-to-beam ratio is nearly independent of the flotation plane, so it isolates shape from datum. Over ±4 mm of heave — a range spanning 13 % of the displaced volume — Sysser 01's ratio stays 0.51 to 0.68 % below the published value and never reaches it, while Sysser 50's spans −0.27 to 0.27 % and passes through −0.03 % at the volume-matched datum. The supplied Sysser 01 surface is therefore a measurably different shape from the one its published row describes, by about 0.6 % in length-to-beam ratio, and the supplied Sysser 50 surface is not.

The consequence for the resistance comparison is a stated bias on one hull and not on the other. It would seem to be the smallest of the four causes, since 0.6 % in a linear dimension is a couple of per cent in wave resistance against the factors of two to four that follow — but section 5.4 finds that ordering contradicted by the resistance comparison itself and withdraws it. Wetted area, the only quantity whose discrepancy has the same sign on both hulls, does not enter the wave resistance at all; it enters the friction line used to reduce the measurement, and there a 0.7 to 2.1 % error in S_c moves the residuary resistance by 3 % of itself at $Fn = 0.30$.

5.2 The running attitude is known, and it brackets rather than fixes the answer

Section 2.8 established that the release fixes the sinkage datum at the centre of gravity and that the pivot follows. The attitude can therefore be applied, and it was: at the pivot the release determines, over the speeds where the sinkage is a large fraction of the draught.

It raises the prediction substantially and by an amount that tracks one quantity. On Sysser 01, at $Fn = 0.40, 0.45, 0.50, 0.55$, the pressure route rises by 59 %, 83 %, 103 %, 74 % while the displaced volume below $z = 0$ rises by 24 %, 42 %, 57 %, 51 %. The resistance rise divided by the volume rise is 1.28, 1.29, 1.29, 1.15, so one quantity accounts for the effect to within 12 % across the range.

Table 7: Effect of imposing the measured running attitude, rotating about the pivot the release determines, against running at the static attitude. The volume ratio is the displaced volume below $z = 0$ at the running attitude over its static value, and it is the quantity that explains the rest of the table.

		sink.	trim	∇/∇_0	R_r	R_W pressure [N]		R_W/R_r	
hull	Fn	[mm]	[°]		[N]	static	running	static	running
Sysser 01	0.40	14.9	−0.56	1.235	6.63	12.36	19.61	1.86	2.96
Sysser 01	0.45	25.0	−1.62	1.420	15.50	6.94	12.72	0.45	0.82
Sysser 01	0.50	30.8	−3.19	1.571	27.98	8.40	17.03	0.30	0.61
Sysser 01	0.55	23.9	−4.67	1.512	38.94	16.79	29.25	0.43	0.75
Sysser 50	0.40	14.8	−0.47	1.279	7.86	11.07	12.04	1.41	1.53
Sysser 50	0.45	19.6	−1.20	1.397	17.41	19.00	25.80	1.09	1.48
Sysser 50	0.50	21.9	−2.11	1.477	29.48	19.02	24.23	0.65	0.82
Sysser 50	0.55	21.0	−2.67	1.482	37.97	17.30	23.67	0.46	0.62
Sysser 50	0.60	19.0	−3.12	1.464	46.02	20.12	28.05	0.44	0.61

The quantity it tracks is the displaced volume, and that identifies the mechanism as the linearisation rather than the attitude. A Neumann–Kelvin solve clips the hull at the undisturbed plane $z = 0$, whereas the real hull at speed sits in the trough of the wave it generates, so the volume below $z = 0$ genuinely exceeds the static value. Imposing the measured sinkage on the clipped hull therefore over-immerses it: the volume rises to 24 to 57 % above static across the speeds run here, and the predicted resistance rises with it, by 59 to 103 %.

The two attitudes therefore bracket rather than bound the answer. The static attitude runs the hull too shallow and under-predicts in the upper range; the measured attitude over-immerses it and over-predicts. On Sysser 01 the bracket is 0.30 to 0.82 of measurement over $Fn = 0.45$ to 0.55. Neither end is the prediction, and a linearised free surface cannot narrow the interval, because doing so requires the wave elevation at the hull — which is what the linearisation has assumed away. Quoting either end alone, as the first version of this report did in quoting the static value, understates the uncertainty by about a factor of two.

One consequence is worth stating plainly, because it inverts an earlier conclusion. The undershoot at high Fn reported in the Results was attributed to running too shallow, and that attribution is confirmed: correcting the attitude removes most of it, taking Sysser 01 from 0.43 to 0.75 of measurement at $Fn = 0.55$. What remains is no longer an attitude error.

5.3 Above $Fn \approx 0.45$ the wetted geometry is incomplete

Both files carry a second patch above the waterline, which the hydrostatics exclude and check for immersion rather than ignore. At the static attitude both clear the free surface — by 40.9 mm on Sysser 50 and 81.5 mm on Sysser 01. At the measured attitude they do not: the excluded patch of Sysser 50 reaches the free surface at $Fn = 0.45$ and is 18.9 mm below it at $Fn = 0.50$

and 35.4 mm at $Fn = 0.65$, and that of Sysser 01 at $Fn = 0.50$ and 28.3 mm at $Fn = 0.60$.

The panel mesh is built from the wetted patch alone, so above those speeds it omits hull surface that is genuinely wetted at the attitude imposed. Sysser 50 crosses first because its after body is wide and shallow. This is a second reason the upper range is not predictable as the code stands, independent of the linearisation argument above, and unlike that one it is fixable: the excluded patch would have to enter the mesh, and the immersion check that currently warns would have to gate the solve rather than the report.

5.4 Beam penalises the low-speed end only, and not the comparison

The remaining discrepancy is discretisation error, and the second hull was chosen to measure its dependence on beam. Sysser 50's beam-to-draught ratio is 6.34 against 3.99, and the oscillation count of a panel pair grows as $|y_i - y_j|/|z_i + z_j|$, so a beamier hull at equal panel count sits closer to the limit of the wave-angle quadrature's validity envelope: the worst count is 366 on the Sysser 50 mesh against 303 on the Sysser 01 mesh at $Fn = 0.30$, with no pair outside the envelope on either.

The expectation was that Sysser 50 would be the worse resolved and the worse predicted throughout. It is neither, and the result is more informative than the expectation.

Table 8: The two hulls compared at the static attitude on the diagnostics that measure discretisation error, split at $Fn = 0.25$. The residual is the root-mean-square body-condition residual at points that are not collocation points, as a fraction of u; the last column is the ratio of the far-field route to the pressure route, which is unity for a converged solution.

hull	N	residual, $Fn < 0.25$	residual, $Fn \geq 0.25$ mean	residual, $Fn \geq 0.25$ max	$Fn \geq 0.25$ R_W/R_r	$Fn \geq 0.25$ far / pressure
Sysser 01	280	0.066–0.147	0.116	0.134	-0.46–+3.09	1.04–3.61
Sysser 50	320	0.166–0.297	0.087	0.095	+0.44–+2.13	0.95–1.79

The beam penalty is real but confined to the bottom of the range. Below $Fn = 0.25$ the body-condition residual on Sysser 50 is 0.166 to 0.297 of u against 0.066 to 0.147 on Sysser 01 — two to four times worse — which is where the wavelength is shortest relative to a panel and where the oscillation count therefore binds. Above $Fn = 0.25$ the ordering reverses on every measure: the mean residual is 0.087 of u on Sysser 50 against 0.116 on Sysser 01, the ratio of prediction to measurement spans 0.44 to 2.13 against -0.46 to 3.09, and the two resistance routes differ by at most a factor of 1.79 against 3.61. Sysser 50 also keeps its predicted resistance positive at every speed, where Sysser 01 returns a negative value at two.

Two conclusions follow, and the second was not anticipated. First, beam is not the discriminator for the quality of the comparison; the resolved wave spectrum is, and it is the low-speed end that is short of it on both hulls. Second, the hull whose geometry reconciles with its published row is also the hull whose resistance reconciles better — which is the opposite of the ordering asserted earlier in this section, where the geometry discrepancy was ranked the smallest of the four causes on the grounds that 0.6 % in a linear dimension is a couple of per cent in resistance. That argument is sound for a small perturbation and it does not account for what is seen here. Two hulls cannot separate geometry fidelity from hull form, so no attribution is made: the observation is recorded, the ranking is withdrawn, and settling it needs a third hull whose file and published row agree and whose form resembles Sysser 01's.

A final observation supports reading the upper range as physics rather than arithmetic. Above $Fn = 0.45$ the two independent resistance routes agree on Sysser 50 to 7 % or better, so discretisation error is demonstrably small there — and the prediction is still only 0.44 to 1.09 of measurement. Where the numerics are converged the model is still short, which points at the causes this section has already named: the omitted immersed surface, the attitude bracket, and the fact that residuary resistance is not wave resistance but contains nonlinear and viscous-form contributions that grow with speed.

6 Capabilities and Limitations

6.1 Capabilities

The code delivers the following on both hulls run here, each verified against something outside itself.

Geometry. It reads an IGES 5.3 file and evaluates the NURBS surface through the Cox–de Boor recurrence, detecting which surface parameter runs around the girth rather than assuming it, checking surface derivatives against a derivative patch built independently from differences of control points, to 1 part in 10^{11}. Hydrostatics converge at second order and are exact to 1 part in 10^{12} on a rectangular box. Plane sections classify vertices lying on the cut plane explicitly, which is not a nicety: with a strict inequality a cut through the pole meridians of a tessellated sphere came out 3.4×10^{-3} low in area against 6.4×10^{-5} for the same cut displaced off the vertices, and waterplanes land on mesh vertices routinely.

Attitude. Sinkage, trim and heel are prescribed as offsets from a reference condition set either by a target displaced volume or by an explicit draught. The geometry is carried as a full hull, so a heeled or asymmetric case is handled without any assumption of port and starboard symmetry, and the two far-field amplitudes $\mathcal{A}_{\pm}$ are kept separately for that reason.

The Kelvin Green function. Both inner quadrature levels are removed analytically, leaving one wave-angle integral, and the resulting evaluation satisfies the free-surface condition to 4×10^{-8} and reaches a median relative accuracy of 5×10^{-15} in the gradient over the panel pairs a Sysser 01 mesh forms at $Fn = 0.30$. The radiation condition's branch is settled by measurement rather than asserted.

The solve. The influence matrix, the constrained solve and both resistance routes reproduce the Michell oracle on a thin Wigley hull to 0.935 and 0.945 of it at 278 panels, agreeing with each other to 1.1 %, as measured here against an independently evaluated oracle. The sphere in an unbounded stream recovers the analytically derived density $\frac{3}{2} u \, n_x$ at first order with a Richardson extrapolant of 1.5008.

Diagnostics. Every solve reports the net source flux, the body-condition residual at points that are not collocation points, the wave-kernel validity envelope and the largest wavelength the mesh can carry. Reporting the residual off the collocation points is the part that matters: at the collocation points the solve makes it vanish by construction, so checking there would be circular.

Refusal. Assembly raises rather than returns when the mesh leaves the quadrature's validity envelope, and the solve raises when the influence matrix is numerically singular. A linear solver will answer a singular system without complaining, and the code declines to pass that answer on.

6.2 Limitations

The limitations fall into three groups: what the model excludes by construction, what the discretisation cannot yet deliver, and what the input data does not determine.

Excluded by the model. The representation (6) is a source distribution and carries no circulation, so it cannot produce lift and therefore cannot represent leeway, appendage lift or induced resistance at any heel angle. Viscosity is absent, so neither frictional nor form resistance is produced and no form factor is implied. The free-surface condition (2) is linear, so wave breaking, spray and finite-amplitude effects are absent, and the hull is clipped at the undisturbed plane $z = 0$ rather than at the wave surface it generates. Depth is infinite and the attitude is prescribed: sinkage and trim are not solved as an equilibrium. The output is the wave resistance of a bare canoe body and nothing else.

Limits of the discretisation. The body-fitted mesh assumed until now that the first surface parameter runs around the girth. That holds for the Sysser 01 file and not for the Sysser 50 file in the same release; the convention is now detected and the patch transposed on loading, but the episode is worth recording because uniform tessellation, and therefore every hydrostatic quantity, was indifferent to it. The density is piecewise constant, which is first order on a curved body, and the wave influences use the centroid rule over the source panel, which carries a bias of order 10 per cent at 158 panels whose sign depends on which resistance route is read. Assembly costs 1164 µs per panel pair, so the influence matrix alone is 28 minutes at 1200 panels with the pressure route's gradient matrix costing the same again; against a ten-minute budget the mesh is capped near 700 panels. That is the binding constraint on the whole exercise, and the mesh sequence available within it does not establish a converged wave resistance for Sysser 01. The far-field route (9) is positive-definite in σ, so its discretisation error is one-sided and it should be read as an upper bound rather than as an estimate. The spectral cut-off can fall below the physical lower limit $\lambda = 1$ at low Froude number, and the reported resistance is then not meaningful even though it is returned.

Limits set by the input data. Three quantities of the supplied Sysser 01 geometry disagree with the published DSYHS hydrostatics by 0.5 to 2.1 %. The repository attributes this to the file being a 2012 re-measurement rather than the towed hull; Sysser 50's file carries the same header and agrees with its own published row to 0.04 % in length and 0.002 % in beam, so the attribution cannot be a systematic property of the release and the discrepancy is hull-specific. It is carried as a stated bias on Sysser 01 and not assumed away. The measured quantity available for comparison is residuary resistance, which is not a measurement but the result of subtracting an ITTC-57 friction line at form factor zero, and therefore contains nonlinear and viscous-form contributions as well as wave resistance. The running attitude is known — the release's Info sheet fixes the sinkage datum at the centre of gravity, and the pivot with it — but it cannot be applied usefully in the upper range, for a reason that belongs to the linearisation rather than to the data: imposing the measured sinkage on a hull clipped at the undisturbed plane $z = 0$ raises the displaced volume by 24 to 57 % above static, because the real hull at speed sits in the trough of its own wave while the model hull does not. The prediction then over-immerses and over-predicts. Above $Fn \approx 0.45$ the running attitude also immerses hull surface that the wetted-surface model excludes, so the geometry itself is incomplete there.

7 Conclusions

The code implements the Neumann–Kelvin problem for a bare canoe body as it states, and the implementation is correct at the level its own oracles can resolve. Its test suite of 183 tests passes. The displaced volume computed by four independent applications of the divergence theorem agrees to 3.0×10^{-15}, and the hydrostatics converge at second order to a Richardson extrapolant of $0.037\,626\,6\,\mathrm{m}^3$. The Kelvin Green function satisfies the linearised free-surface condition to 3.8×10^{-8}, a property the code never imposes numerically and so an independent check on the whole derivation, and its gradient reaches a median relative accuracy of 5.3×10^{-15} over the panel pairs a Sysser 01 mesh forms at $Fn = 0.30$.

The wave resistance of Sysser 01 is not converged on any mesh the code's throughput permits. Across the eleven measured speeds the ratio of prediction to measurement spans −0.46 to 4.41 by the pressure route, and the predicted curve peaks at $Fn = 0.60$ while the measurement rises monotonically. Refining from 280 to 480 panels at $Fn = 0.30$ moves the pressure value by only +0.9 % and halves the body-condition residual, so the solve is converging even where the resistance is not, and the resistance constant $\rho g^2/(\pi u^4)$ is confirmed end to end against an independently evaluated Michell integral. What is not established is a wave resistance for this hull that could be used in place of the series regression.

Running the second hull identified where the discrepancies come from, and they are four separable things rather than one. The geometry discrepancy is specific to Sysser 01, whose supplied surface differs from its published row by 0.6 % in a quantity that no flotation plane can adjust, while Sysser 50's agrees to 0.03 %; it is also the smallest of the four. The running attitude is determined by the release and not, as previously recorded, undetermined, but imposing it on a hull clipped at the undisturbed plane over-immerses that hull, so the two attitudes bracket the answer at about a factor of two and the linearisation cannot do better. Above $Fn \approx 0.45$ the running attitude immerses hull surface the wetted-surface model excludes, so the geometry is incomplete there. And the largest cause is discretisation error, which the second hull measures against beam: at comparable panel counts the beamier hull carries the larger body-condition residual, on the diagnostic the solve never sees.

Three conclusions follow for anyone using the code. Quote the pressure route, with the far-field value beside it as an upper bound, because the far-field integral is positive-definite in the source density and its discretisation error is one-sided. Read the body-condition residual measured away from the collocation points, not the net source flux, which the constrained solve drives to machine zero and which therefore carries no information once the constraint is on. And treat the comparison with residuary resistance as a comparison with a proxy: the measured quantity is a total resistance minus an ITTC-57 friction line at form factor zero, so it contains nonlinear and viscous-form contributions that the model does not produce.

Two improvements would change the picture, and neither is a larger mesh. Tabulating and interpolating the Kelvin kernel would recover the factor of ten in throughput that a converged mesh needs, and pointwise accuracy is now established well enough for that to be safe. Replacing the piecewise-constant density with a linear one would address the first-order convergence that is the actual cause rather than its cost. Independently of either, the upper range now needs two things that are not mesh refinement: the excluded patch brought into the panel mesh, so that the wetted geometry at the running attitude is complete; and a free-surface condition applied at the wave elevation rather than at $z = 0$, without which the attitude bracket cannot be narrowed.

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